MURRAY STATE UNIVERSITY · BRYANT HARRISON

Wavelet Lab — Teacher Guide

Haar representations, coefficient selection, and numerical reconstruction error

WHAT STUDENTS LEARN

Wavelet Lab is a short, hands-on introduction to **orthonormal representation**. It begins with eight values, where students can see exactly what each Haar coefficient does, then carries the same idea into a 64 × 64 procedural grayscale scene. The teaching sequence is **predict** → **try** → **explain** → **new case**: students predict a coarse-only reconstruction, toggle details, explain the resulting local changes, and transfer the idea to an image budget.

Open `index.html` in any current browser. It is one self-contained offline page; no login, server, image download, or runtime inference is involved. Use **?** to reopen the student how-to and ⚙ for Presentation mode and presenter notes.

THE EXACT MATH

For every neighboring pair `(a,b)`, the lab calculates the normalized Haar coarse coefficient `(a+b)/√2` and detail `(a−b)/√2`. It repeats this only on the coarse portion: an 8-value vector is laid out as `[A3, D3, D2.1, D2.2, D1.1 … D1.4]`. The two-dimensional transform applies that same operation to rows and columns of each successively smaller top-left block.

> **Why normalized?** The transform preserves sum of squares: energy in the data equals energy in its coefficients. That gives this lab an exact check: when coefficients are set to zero, the energy omitted equals the squared reconstruction error.

20-MINUTE RUN OF SHOW

1. **0–3 min: predict.** Project the default signal. Ask what "coarse only" retains before pressing it. At full coarseness, its inverse reconstruction makes every signal value the global average; the stored coefficient itself is normalized.
2. **3–7 min: try.** Toggle the D1 cards one by one. Students should identify each affected adjacent pair before you reveal that D1 values correct the finest pairwise differences. Then toggle a coarser detail and ask which broader region changes.
3. **7–12 min: explain.** Change a source slider, then compare coefficient values. Ask which scale carries the new feature. Use the signal's numerical squared error and omitted energy to anchor the visual impression in a checkable quantity.
4. **12–17 min: image.** Set the Mountain lake to 64 coefficients, then 256 and 1,024. Ask when the large regions return, when boundaries return, and what still looks wrong.
5. **17–20 min: new case.** Switch to Portrait at 64. Students name one broad feature that persists and one detail that fails, then raise the slider until eyes become recognizable.

WHAT THE IMAGE CONTROL MEANS

The slider retains exactly the displayed number of **largest absolute-value coefficients**; ties break by fixed array order. Each selected coefficient remains at its original floating-point value and every other coefficient is zero before inverse transformation. The count is a coefficient budget, not bytes, a bit rate, a compression ratio, or a claim about a saved file.

For an orthonormal transform, choosing the largest magnitudes is the best k-term selection for squared error. It does **not** guarantee perceptual quality: a small coefficient that supports a sharp edge can still matter visibly. The error canvas brightens absolute pixel error threefold for visibility and then clips to gray values. The displayed RMSE is computed from the unclipped reconstruction.

MISCONCEPTIONS WORTH CATCHING

- **"Coarsening means information is destroyed."** The normalized coarse coefficient plus the detail reconstruct the pair exactly. Loss begins only when a coefficient is omitted.
- **"Blur is denoising."** This lab does not model noise or choose a denoising threshold. It demonstrates a fixed coefficient-selection reconstruction.
- **"A coefficient budget is compression."** It is a precondition for representation sparsity, not a complete file format. Real compression also needs quantization and coding.
- **"More retained energy always looks better."** Energy and perceived quality are related but not identical objectives.

ASSESSMENT PROMPTS

1. For `[3,1]`, calculate the normalized coarse coefficient and detail. If only the coarse coefficient remains, why does reconstruction become `[2,2]` and why is its squared error 2?
2. Which wavelet scale would record a change confined to one adjacent pair? Which carries a broad change across the whole signal?
3. At the same coefficient budget, why might a portrait and a city look differently faithful?
4. Ask students to defend or reject: "the image with lower RMSE must look better."

METHODS AND ATTRIBUTION

The implementation is a didactic Haar transform, independently tested against a small explicit Haar matrix for the 8-value case; it also checks round trips, orthogonality, energy preservation, omitted-energy/squared-error equality, and exhaustive small best-k subsets. The procedural scenes are original illustrative fixtures, not photographs.

Further reading: MathWorks, Wavelet Data Compression (decompose, threshold, reconstruct; coefficient energy) and Two-Dimensional Wavelet Packet Analysis (2-D context and thresholding). These links are reading sources, not runtime dependencies; the lab does not claim an assignment, endorsement, or current course use.